MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special

Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Value Addition Course(VAC) - MG3VACCSC200 - WHITE HAT HACKING

(2024 ADMISSION ONWARDS)

Part A

Q1. What does an exploit take advantage of?

An exploit takes advantage of a security vulnerability or weakness in software, hardware, or a network.

Q2. What is information gathering in ethical hacking?

Information gathering (reconnaissance) is the process of collecting information about a target before testing.

Q3. List two risks caused by eavesdropping attacks.

1. Theft of confidential information.
2. Identity theft or unauthorized access.

Q4. Identify the primary goal of an ethical hacker related to TOE.

To identify vulnerabilities in the Target of Evaluation (TOE) and recommend security improvements.

Q5. Summarize how firewalls can affect the result of a ping sweep.

Firewalls can block ICMP Echo Requests and Replies, causing active hosts to appear offline.

Part B

Q6. Discuss how a hacker can use an attack to exploit a known vulnerability.

A hacker identifies the target, scans for vulnerabilities, uses a known exploit, gains access, escalates privileges, and maintains access. Prevention includes patching, updates, firewalls, and antivirus.

Q7. Differentiate between manual exploitation and scripted exploitation.

Manual exploitation is human-driven and slower but more accurate. Scripted exploitation uses automated tools, is faster, and suitable for large networks.

Q8. Explain the concept of password spraying in remote password guessing.

Password spraying involves trying one common password against many user accounts to avoid account lockout. Prevention: MFA, strong passwords, and lockout policies.

Q9. Explain the importance of Confidentiality, Integrity, and Availability (CIA).

Confidentiality protects data from unauthorized access. Integrity ensures data accuracy. Availability ensures systems and data are accessible when needed.

Q10. Discuss reconnaissance phase.

Reconnaissance is the first phase of ethical hacking involving passive and active information gathering such as IP addresses, domains, operating systems, ports, and services.

Q11. Explain how human errors can act as threats to information security.

Weak passwords, phishing, misconfiguration, sharing confidential data, and ignoring updates can lead to security breaches.

Q12. Compare NESSUS and NMAP in port scanning.

Nmap is a network and port scanning tool. Nessus is a vulnerability assessment tool that detects security weaknesses.

Nmap	**Nessus**
Network scanning tool	Vulnerability scanner

Nmap	Nessus
Finds open ports and services	Detects vulnerabilities and misconfigurations
Fast and lightweight	More comprehensive
Mainly used for reconnaissance	Used for vulnerability assessment
Open-source	Commercial (with Essentials free version)

Part C

Q13. Explain port scanning and its types.

Port scanning is the process of checking network ports to identify open ports and services running on a target system.

Types of Port Scanning:

1. TCP Connect Scan
2. SYN (Half-Open) Scan
3. UDP Scan
4. FIN Scan
5. NULL Scan
6. Xmas Scan
7. ACK Scan
8. Idle Scan

Uses:

- Identify open ports.
- Detect running services.
- Find security weaknesses.
- Assist in penetration testing.

Advantages:

- Improves security.
- Helps administrators detect vulnerabilities.
- Supports network auditing.

Q14. Explain offline password cracking techniques in detail.

Offline password cracking is the process of cracking password hashes without interacting with the target system.

Techniques:

1. Dictionary Attack
2. Brute Force Attack
3. Hybrid Attack
4. Rainbow Table Attack
5. Mask Attack
6. Rule-based Attack

Tools:

- John the Ripper
- Hashcat
- Ophcrack

Advantages:

- No risk of account lockout.
- Faster than online attacks.
- Can use GPU acceleration.

Prevention:

- Use strong, unique passwords.
- Enable Multi-Factor Authentication.
- Use salted password hashes.
- Implement secure password policies.
- Use modern hashing algorithms such as **bcrypt**, **scrypt**, or **Argon2**.